

Earthquake Magnitude Prediction Using Machine Learning

1. Project Overview

Objective

The objective of this project is to build a machine learning model capable of predicting earthquake magnitude using seismic and geographical features.

This project uses ensemble machine learning algorithms such as Random Forest and XGBoost to analyze earthquake-related data and estimate earthquake magnitude.

2. Problem Statement

Earthquakes are natural disasters that can cause severe damage to infrastructure and human life. Predicting earthquake magnitude can help improve disaster preparedness and risk assessment.

This project focuses on developing a regression-based machine learning model that predicts earthquake magnitude using historical seismic data.

3. Dataset Information

Dataset Features

The dataset contains earthquake-related information such as:

Numerical Features

- latitude
- longitude
- depth
- nst
- gap
- rms
- horizontalError
- depthError

Categorical Features

- magType

- locationSource
- magSource
- country
- depth_category

Time-Based Features

- year
- month
- day
- hour

Target Variable

- mag (Earthquake Magnitude)
-

4. Exploratory Data Analysis (EDA)

Several exploratory data analysis techniques were performed to better understand the dataset.

EDA Steps

1. Dataset Inspection

- Checked dataset shape
- Inspected column names
- Verified data types
- Identified duplicate rows

2. Missing Value Analysis

- Checked missing values using:

3. Target Distribution Analysis

- Visualized earthquake magnitude distribution
- Checked skewness
- Analyzed mean and median

4. Correlation Analysis

- Used correlation heatmap for numerical features
- Identified highly correlated features

5. Outlier Detection

- Used boxplots and IQR method
 - Investigated extreme earthquake values
-

5. Data Preprocessing

Several preprocessing techniques were applied before model training.

Steps Performed

1. Handling Missing Values

Missing values were handled appropriately using:

- Mean imputation
- Median imputation
- Column removal (if necessary)

2. Feature Engineering

Custom features were created to improve model performance.

Country Extraction

Country names were extracted from the `place` column.

Time Feature Extraction

The `time` feature was converted into:

- year
- month
- day
- hours

Depth Categorization

Earthquake depth was categorized into:

- Shallow
- Intermediate
- Deep

3. Encoding

Categorical features were encoded using:

- One-Hot Encoding

4. Outlier Handling

Outliers were analyzed using:

- IQR Method

- Boxplots

5. Feature Selection

Feature importance analysis was performed to remove noisy or weak features.

6. Train-Test Split

A stratified train-test split approach was used to maintain balanced target distribution.

7. Machine Learning Models Used

Multiple regression algorithms were tested.

Models Explored

- Linear Regression
 - Decision Tree Regressor
 - Random Forest Regressor
 - Gradient Boosting Regressor
 - XGBoost Regressor
-

8. Final Model Selection

After comparing multiple models, XGBoost and Random Forest produced the best results.

Final Selected Model

XGBoost Regressor

Reasons for selection:

- Handles nonlinear relationships effectively
 - Performs well on structured/tabular data
 - Robust against overfitting after tuning
 - Good generalization performance
-

9. Hyperparameter Tuning

Hyperparameter tuning was performed using `RandomizedSearchCV`.

Parameters Tuned

Random Forest

- n_estimators
- max_depth
- min_samples_split
- min_samples_leaf
- max_features

XGBoost

- learning_rate
 - max_depth
 - n_estimators
 - subsample
 - colsample_bytree
-

10. Model Evaluation Metrics

Several regression metrics were used to evaluate performance.

R² Score

Measures how well the model explains variance.

Mean Absolute Error (MAE)

Measures average absolute prediction error.

Root Mean Squared Error (RMSE)

Measures overall prediction error.

11. Final Model Performance

XGBoost Performance

Training Metrics

- R² Score: 0.7206
- MAE: 0.1719
- RMSE: 0.2277

Testing Metrics

- R² Score: 0.6154

- MAE: 0.2012
- RMSE: 0.2629

Interpretation

The model demonstrates:

- Good predictive capability
- Stable generalization
- Controlled overfitting
- Effective feature engineering

Considering earthquake prediction is a noisy and highly complex real-world problem, this performance is meaningful and practically useful.

12. Feature Importance Analysis

Feature importance analysis was performed to identify the most influential variables.

Top Important Features

- nst
- gap
- year
- country
- depth
- latitude
- longitude

Insights

- Seismic station count (nst) strongly influences predictions.
- Geographical location contributes significantly.
- Temporal patterns also affect earthquake magnitude.

13. Cross Validation

Cross-validation was used to verify model stability.

Technique Used

- K-Fold Cross Validation

Benefits:

- Reduces evaluation bias
- Measures model consistency

- Improves reliability
-

14. Visualizations Included

The project includes several important visualizations:

- Target Distribution Plot
 - Correlation Heatmap
 - Residual Plot
 - Actual vs Predicted Plot
 - Feature Importance Graph
-

15. Deployment

The trained machine learning model deployed using:

- Streamlit

17. Saving the Model

18. Future Improvements

Possible future enhancements:

- **Larger datasets**
 - Real-time seismic data integration
 - Deep learning models
 - Geospatial modeling
 - Advanced feature engineering
 - Improved deployment dashboard
-

19. Conclusion

This project successfully demonstrates the application of machine learning techniques for earthquake magnitude prediction.

The workflow included:

- Data preprocessing
- Feature engineering
- Model training
- Hyperparameter tuning

- Explainability analysis
- Performance evaluation

The final XGBoost model achieved strong and stable performance on a noisy real-world dataset.

This project highlights the importance of:

- feature engineering
- proper validation
- explainable AI
- model generalization

in real-world machine learning systems.

20. Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-learn
 - XGBoost
 - Joblib
-

21. Author

Sourabh

Machine Learning Project